

CLGNN: A Contrastive Learning-based GNN for Temporal Betweenness Prediction under Extreme Value Imbalance

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Abstract

Temporal Betweenness Centrality (TBC) measures how often a node appears on optimal temporal paths, reflecting its importance in temporal networks. However, exact computation is highly expensive, and real-world TBC distributions are extremely imbalanced, causing learning-based models to overfit to zero-centrality nodes and fail to identify truly central nodes. Existing graph neural networks (GNNs) either ignore temporal dependencies or cannot handle such extreme imbalance. To address these issues, we propose CLGNN, a scalable and inductive contrastive learning-based GNN for accurate TBC prediction. CLGNN preserves temporal path validity through an instance graph and encodes structural, path-time aware dependencies via dual aggregation. To mitigate imbalance, a stability-based clustering-guided contrastive module separates nodes of different centrality levels in representation space, while a regression head estimates TBC values. Extensive experiments on diverse benchmarks demonstrate that CLGNN is scalable, generalizable, and effective.

CCS Concepts

• Theory of computation → Graph algorithms analysis; • Computing methodologies → Neural networks.

Keywords

Temporal Graph; Betweenness Centrality; Graph Neural Networks; Contrastive Learning

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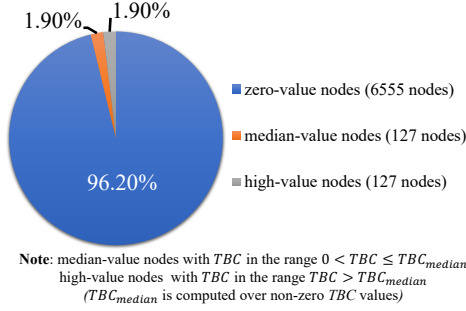
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1 Introduction

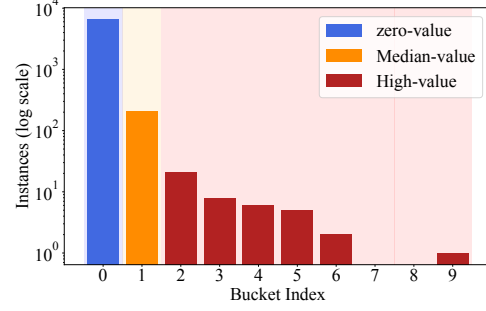
Real-world systems such as social and communication networks are often modeled as temporal graphs, where edges are associated with time intervals or timestamps, indicating when they are active. In such graphs, certain individuals play a crucial role in driving information diffusion or facilitating rumor spreading. Temporal generalizations of betweenness centrality are defined based on the optimal temporal paths. However, a significant challenge lies in the computational complexity of these generalizations, as calculation requires evaluating all possible temporal paths, which is resource-intensive. Existing graph learning models primarily focus on either computing betweenness centrality (BC) for static graphs [10, 20] or performing vertex ranking on temporal graphs [19, 43]. To the best of our knowledge, only DBGNN [15] predicts TBC values on temporal graphs. However, its prediction accuracy is low, our experiments show that the mean absolute error (MAE) for TBC reaches 496 on the dataset *Highschool2013*. Unlike conventional ranking tasks, predicting TBC values on temporal graphs is more challenging due to the extreme class imbalance problem.

To illustrate this, we conducted a statistical analysis of the distribution of TBC values in the *ia-reality-call* dataset, as shown in Figure 1. Figure 1a groups nodes into zero-, median-, and high-value categories, revealing a highly imbalanced distribution, with 96.2% of nodes having zero TBC, while only 1.9% fall into each of the median and high-value categories. Figure 1b presents a log-scale histogram of TBC values. The non-zero range is evenly divided into 9 equal-width intervals, with a separate bucket for zero values. It further demonstrates that high TBC nodes are not only rare but also span a wide range of magnitudes. This extreme imbalance causes two problems: (1) **Training is inefficient as most TBC values of vertices are zero, which contribute no useful learning for critical nodes with large TBC values;** (2) **the zero values can overwhelm training and lead to degenerate models.**

In such circumstances, applying existing temporal graph models (e.g., TATKC [43], DBGNN [15]) results in significant limitations. These models fail to adequately learn the true TBC values of vertices in the median and high-value intervals, leading to most predicted



(a) The proportion of nodes in 3 categories



(b) The distribution of TBC values across 10 buckets

Figure 1: Distribution of TBC values in the *ia-reality-call* dataset

TBC values in these categories being erroneously classified as 0. This is particularly problematic as vertices in the median and high-value intervals are critical hubs for analyzing information flow and propagation. **We identify TBC distribution imbalance during training as the main obstacle impeding the model from achieving good accuracy.**

Directly applying techniques from imbalanced text and image learning to graph data, such as oversampling, undersampling, loss re-weighting, and data augmentation (e.g., node dropping, edge perturbation, and feature masking), has proven ineffective, as demonstrated in the experimental results shown in Table 6. Motivated by contrastive learning, which pulls together representations of similar instances and pushes apart dissimilar ones, we adopt this paradigm for TBC prediction. Our objective is to improve the prediction of medium and high TBC values, particularly by avoiding their misclassification as zero-value nodes. Concretely, contrastive learning makes zero-value vertices more compact in representation space and increases their separation from medium/high-value nodes, enabling better discrimination of their distinctive patterns and improving predictions across all TBC ranges.

Based on these insights, we propose CLGNN, a Contrastive Learning-based GNN with a representation module and a prediction module. The representation module captures path-based and temporal dependencies via a dual aggregation mechanism that integrates mean aggregation with edge-to-node multi-head attention, enhanced by temporal path count and time encodings. The prediction module contains two complementary components, KContrastNet and ValueNet. KContrastNet introduces a stability-aware clustering strategy to guide contrastive learning. It first performs stability-screened clustering to form stable and structurally coherent groups, and then samples contrastive pairs within each cluster. Pairs are further reweighted by their value gaps, aligning close-valued nodes while enforcing stronger margins for distant ones. This strategy separates low/medium/high-centrality nodes in representation space and captures subtle temporal-importance differences among structurally similar nodes, enabling accurate identification of rare but influential vertices. ValueNet uses multi-layer perceptrons (MLPs) to estimate TBC values. To summarize, our main contributions are as follows:

- We first propose an inductive and scalable contrastive learning-based TBC regression model, trained on small graphs yet generalizable to unseen massive graphs with varying size, skewness,

and different definitions of temporal paths and centrality metrics. In contrast to the state-of-the-art method DBGNN [15], which trains and tests on the same graph and thereby limits its generality, our model supports cross-graph inference.

- The proposed model, CLGNN, jointly learns path-time dual-aware node representations and refines node embeddings via an improved stability-based clustering-guided contrastive learning to enhance TBC prediction.
- Extensive experiments show that CLGNN significantly outperforms state-of-the-art static and temporal GNNs as well as both exact and approximate TBC computation methods. It achieves up to $663.7\times$ speedup over exact computation, while reducing MAE by up to $26.36\times$ and improving Spearman correlation by up to $16.50\times$ compared to leading static GNNs. Against the strongest temporal GNNs, CLGNN attains up to $8.44\times$ lower MAE and $3.64\times$ higher correlation, and under the same runtime budget, it outperforms the best approximate method with up to $267.90\times$ lower MAE and $2.75\times$ higher correlation, while avoiding failures on medium-to-massive graphs.

2 Related Work

Traditional BC computation method. Representatives for exact BC computation on static graphs include Brandes' algorithm [1] and its variants, such as Brandes++ [9] and BADIOs [30]. Approximate methods leveraging techniques such as source sampling [2] and node-pair sampling [7] to improve accuracy and efficiency. In temporal graphs, research has focused on temporal paths [4] and BC variants. Tsalouchidou et al. [34] explored exact computation using static snapshots or predecessor graphs. Methods like ONBRA [29] have introduced approximation techniques with probabilistic guarantees. Zhang et al. [44] proposed exact and approximate methods based on transformed time instance graphs. Regunta et al. [25] leveraged parallelism to update TBC values incrementally. Although the above-mentioned optimization techniques have been proposed, the computational complexity of TBC is still relatively high for large-scale temporal graphs.

Graph neural network based model. Dynamic graph modeling focuses on capturing evolving structures and enabling scalable temporal representation learning [6, 38–41]. Models such as TGAT [37], TGN [26], and GATv2 [3] leverage self-attention to model temporal dependencies, with extensions to heterogeneous graphs via

cross-type homophily [31]. De Bruijn Neural Networks [24] extend causality-aware mechanisms for temporal graphs. Temporal Transformer [36] models dynamic graphs as sequences, utilizing attention mechanisms to capture temporal dependencies. TempGNTK [33] offers a theoretical kernel-based framework for temporal graph classification. Nevertheless, these models failed to capture the intricate path structures required for betweenness computations, leading to low estimation accuracy. Some models are specially designed for BC ranking or value prediction on static graphs. Recent approaches [10, 21] exploit graph attention and reinforcement learning to approximate centrality. In temporal settings, TATKC [43] uses time-injected attention for Katz centrality ranking, and DBGNN [15] predicts centrality with a time-aware GNN. While effective in ranking, these models struggle with precise TBC value prediction, and they often misclassify mid-to-high value nodes as zero. Additionally, DBGNN is transductive, limiting generalization to unseen networks.

3 Preliminaries

Definition 1. (Temporal Graph). A directed temporal graph is denoted by $G = (V, E, T)$, where (i) V is a vertex set; (ii) $E \in V \times V \times T$ is the set of directed time-stamped edges; and (iii) T is the set of discrete or continuous time points at which interactions occur. Each edge $e = (u, v, t) \in E$ represents a connection from node u to v in V at a specific time $t \in T$.

Definition 2. (Temporal Path). A temporal path in $G = (V, E, T)$ is a sequence of edges $p = u \xrightarrow{t_1} w_1 \cdots \xrightarrow{t_{m-1}} w_{m-1} \xrightarrow{t_m} v$, where (i) each edge in p belongs to the edge set E ; (ii) the sequence of timestamps satisfied $0 < t_{i+1} - t_i \leq \delta$ for $1 \leq i < m$, where $\delta \in \mathbb{R}$, ensuring that the interactions not only follow an ascending temporal order but also occur within a bounded time interval.

Temporal graphs differ from static graphs due to temporal constraints, which yield multiple notions of optimal paths, such as shortest paths (minimizing the number of hops), earliest-arrival paths (reaching the destination as early as possible), and latest-departure paths (departing as late as possible while still arriving on time), among others. Based on optimal paths, the formal definition of temporal betweenness centrality (TBC) [34, 44] is as follows:

Definition 3. (Temporal Betweenness Centrality, TBC). Given a temporal graph $G = (V, E, T)$, the normalized temporal betweenness centrality value $TBC(v)$ of a node $v \in V$ is defined as:

$$TBC(v) = \frac{1}{|V|(|V| - 1)} \sum_{\forall s \neq v \neq z \in V} \frac{\sigma_{sz}(v)}{\sigma_{sz}},$$

where $|V|$ is the total number of vertices in G ; σ_{sz} denotes the number of optimal temporal paths from s to z ; and $\sigma_{sz}(v)$ represents the number of such paths pass through v .

4 Our Method

To effectively predict TBC under strong data imbalance, we propose a contrastive learning-based GNN model, CLGNN, as illustrated in Figure 2. CLGNN is composed of two main components, i.e., a representation module and a prediction module.

4.1 Representation Module

To capture temporal paths and preserve temporal dependencies, the temporal graph is transformed into an instance graph, where each node v is associated with the set $T_{\text{out}}(v)$ of timestamps corresponding to its outgoing edges. For each node u and its outgoing edge (u, v, t) , the number of valid temporal paths is defined as:

$$P(u, v, t) = \sum_{t_v \in T_{\text{out}}(v)} \mathbb{I}(t_v - t),$$

where $\mathbb{I}(\cdot)$ denotes the indicator function given by

$$\mathbb{I}(x) = \begin{cases} 1, & \text{if } x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

If $P(u, v, t) = 0$, then the message generated by node u will not be transmitted to node v through the edge with timestamp t during the message passing process.

Path-Time dual-aware mechanism. It combines continuous time encoding and temporal path count encoding, jointly capturing temporal dependencies and relevant path information in the message passing process.

• **Continuous Time Encoding.** We apply Bochner’s theorem and Monte Carlo integral [37] to map a temporal domain $T = [0, t_{\max}]$ to the d -dimensional vector space.

$$\Phi_{\text{time}}(t) = \sqrt{\frac{1}{d_T}} [\cos(\omega_1 t), \sin(\omega_1 t), \dots, \cos(\omega_{d_T} t), \sin(\omega_{d_T} t)]^T,$$

where d_T is the finite dimension and $\omega = (\omega_1, \dots, \omega_{d_T})^T$ are learnable parameters.

• **Temporal Path Count Encoding.** We propose a path-count based strategy, which processes the number of temporal paths through a logarithmic transformation and neural projection to provide stable and efficient input features for the network.

$$P(u, v, t) \xrightarrow{\text{log-transformed value}} \log(1 + P(u, v, t)),$$

$$\Phi_{\text{path}}(u, v, t) = \text{MLP}(\text{reshape}(\log(1 + P(u, v, t)))),$$

where the reshaping operation $\text{reshape}(1, 1)$ converts the scalar into a suitable format for input into the network. Note that the path count $P(u, v, t)$ undergoes a $\log(1 + P)$ transformation, which aims to compress the influence of large path values on model training, alleviating gradient explosion issues while preserving non-negativity and the distinguishability of the values.

Message Function. For each interaction event $e = (u, v, t)$, a message is computed to update the representation of node v . Specifically, the message at layer l is defined as:

$$m_{uv}^l(t) = \left(h_u^{l-1}(t^-) || h_v^{l-1}(t^-) || \Phi_{\text{time}}(t - t^-) || \Phi_{\text{path}}(u, v, t) \right),$$

where $h_u^{l-1}(t^-)$, $h_v^{l-1}(t^-)$ are the hidden representation for node u and v in the previous layer just before time t , and $||$ denotes concatenation. The computed message $m_{uv}^l(t)$ is then passed into the aggregation function and contributes to the update of node v ’s embedding at layer l .

Dual Message Aggregator Mechanism. Since each node typically interacts with multiple neighbors and may receive multiple messages from the same neighbor at different times, there exist two levels of message aggregation: across edges and across nodes. To address this, we employ a dual aggregation mechanism, i.e., mean

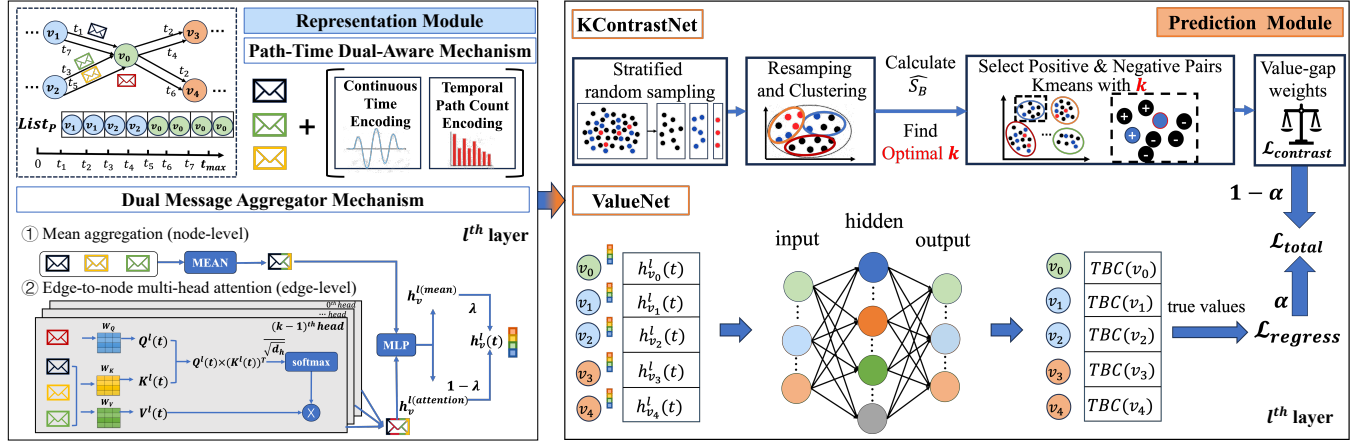


Figure 2: Overview of the Proposed Model: CLGNN

aggregation and edge-to-node multi-head attention, to balance the model's stability and flexibility.

• **Mean aggregation (node-level).** Mean aggregation performs aggregation at the node level, ensuring that all neighbors contribute equally to the target node's representation. At the l -th layer, the mean-aggregated message received by v at time t is computed as:

$$\bar{h}_v^l(t) = \frac{1}{|N_{in}(v)|} \sum_{u \in N_{in}(v)} \left(\sum_{t_x \in T_{uv}} \frac{P(u, v, t_x)}{\sum_{t'_x \in T_{uv}} P(u, v, t'_x)} \cdot m_{uv}^l(t_x) \right),$$

where $N_{in}(v)$ is the set of v 's in-neighbors; T_{uv} is the set of time-stamps associated with all edges from u to v . Notably, only temporal paths that are reachable (i.e., those with $P(u, v, t_x) > 0$) are considered in the aggregation.

• **Edge-to-node multi-head attention (edge-level).** This mechanism computes attention for each edge connected to a node, ensuring that all temporal edges contribute to the node's representation update. The input matrix of the l -th attention layer is:

$$M_v^l(t) = [m_{uv}^l(t_x)]_{\forall (u, v, t_x) \in E_{in}(v), P(u, v, t_x) > 0}^T,$$

$E_{in}(v)$ is the set of v 's incoming edges. Then $M^l(t)$ is projected into query, key, and value spaces: $Q^l(t) = [h_v^{l-1}(t^-) \parallel h_v^{l-1}(t^-) \parallel \Phi_{\text{time}}(t - t^-) \parallel \Phi_{\text{path}}(u, v, t)] \cdot W_Q$; $K^l(t) = M^l(t) \cdot W_K$; $V^l(t) = M^l(t) \cdot W_V$, where $W_Q, W_K, W_V \in \mathbb{R}^{(d+d_T) \times d_h}$ are learnable matrices.

Next, the scaled dot-product attention is used in attention layers. The aggregated message for node v at the l -th layer, computed via the multi-head attention mechanism, is given by:

$$\begin{aligned} \tilde{h}_v^l(t) &= \text{Attn}(Q^l(t), K^l(t), V^l(t)) \\ &= \text{softmax}\left(\frac{Q^l(t)(K^l(t))^T}{\sqrt{d_h}}\right) V^l(t) \in \mathbb{R}^{d_h}. \end{aligned}$$

Embedding. The final embedding of node v at time t , denoted as $h_v^l(t)$, is obtained by combining the outputs of mean aggregation

and edge-to-node multi-head attention:

$$\begin{aligned} h_v^l(t) &= \lambda \cdot h_v^{l(\text{mean})}(t) + (1 - \lambda) \cdot h_v^{l(\text{attention})}(t), \\ h_v^{l(\text{mean})}(t) &= \text{MLP}(\bar{h}_v^l(t) \parallel h_v^{l-1}(t)), \\ h_v^{l(\text{attention})}(t) &= \text{MLP}(\tilde{h}_v^{l(0)}(t) \parallel \tilde{h}_v^{l(1)}(t) \parallel \dots \parallel \tilde{h}_v^{l(k-1)}(t) \parallel h_v^{l-1}(t)), \end{aligned}$$

where k is the number of heads. λ balances the contribution of the two aggregation types.

4.2 Prediction Module

To address the imbalance in temporal betweenness centrality values and enhance the model's sensitivity to rare but important nodes, we design two complementary components, i.e., KContrastNet and ValueNet, in the prediction module.

KContrastNet. KContrastNet employs a contrastive learning strategy to effectively distinguish features among low-, medium-, and high-value nodes in TBC prediction. To enhance the selection of positive and negative sample pairs, we first introduce a stability-based clustering approach that identifies the optimal number k of clusters in the node representation space.

• **Stability-based clustering method.** Traditional methods for choosing the number of clusters, such as the elbow method [32] or silhouette score [28], often struggle with noisy data or class imbalance, leading to unstable results. To address this, we propose a stability-based clustering method that combines bootstrap resampling [11] with K-means clustering [14] to estimate the optimal number of clusters. To improve efficiency, we first perform stratified random sampling on the original dataset F to obtain a reduced subset F^n (e.g., 40% of the original size) that retains the original distribution. Bootstrap resampling is then applied on F^n to generate B independent sample pairs $\{(X_i, Y_i)\}_{i=1}^B$. For each pair and each candidate cluster number $k \in \{2, \dots, K\}$, we run k-means separately on X_i and Y_i to obtain two cluster assignments ψ_{X_i} and ψ_{Y_i} , respectively. We then compute the clustering distance between these assignments and average the results over all B pairs to obtain a clustering instability, as formally defined below. Algorithmic details and complexity analysis for the stability-based clustering are provided in Appendix A.1.

Definition 4. (Clustering Distance). The clustering distance between two clustering results ψ_{X_i} and ψ_{Y_i} is defined as:

$$d(\psi_{X_i}, \psi_{Y_i}) = E_{x, y \in (X_i \cap Y_i)} [\delta(\psi_{X_i}(x), \psi_{X_i}(y)) - \delta(\psi_{Y_i}(x), \psi_{Y_i}(y))],$$

where E denotes the expectation over all samples x and y . $\psi_{X_i}(x)$ denotes the cluster label assigned to sample x in ψ_{X_i} . The function

$$\delta(a, b) \text{ is the Kronecker Delta, defined as: } \delta(a, b) = \begin{cases} 1, & \text{if } a = b, \\ 0, & \text{if } a \neq b. \end{cases}$$

Definition 5. (K-means Clustering Instability). The clustering instability quantifies the variation in clustering results across different resamples and is defined as:

$$\hat{s}_B(\psi, k, n) = \frac{1}{B} \sum_{i=1}^B d(\text{KMeans}(X_i, k), \text{KMeans}(Y_i, k)),$$

ψ is the clustering results of KMeans function and the optimal number of clusters $\hat{k} = \arg \min_{2 \leq k \leq K} \hat{s}_B(\psi, k, n)$ is obtained by minimizing the instability. The complete algorithm and its complexity analysis are provided in Appendix A.1.

• **Contrastive Learning.** After clustering, each node obtains a cluster label. To ensure semantic and structural consistency, supervised contrastive pairs are restricted within each cluster. Positive pairs are selected from in-cluster nodes with *small* TBC gaps, while negative pairs come from nodes with *large* TBC gaps.

In-cluster candidate sets. The positive and negative candidate sets are defined as:

$$\mathcal{S}^+(u) = \{v \mid \begin{matrix} \delta(\psi(u), \psi(v))=1 \\ 0 < |\Delta_{uv}| \leq \gamma_{\text{pos}} TBC_{\text{median}} \end{matrix} \}, \mathcal{S}^-(u) = \{v \mid \begin{matrix} \delta(\psi(u), \psi(v))=1 \\ |\Delta_{uv}| \geq \gamma_{\text{neg}} TBC_{\text{median}} \end{matrix} \},$$

where $\Delta_{uv} = TBC(u) - TBC(v)$ and thresholds $\gamma_{\text{pos}}, \gamma_{\text{neg}} \in (0, 1)$ with $\gamma_{\text{neg}} > \gamma_{\text{pos}}$. The median provides robustness against heavy-tailed distributions.

Value-gap weights. To emphasize informative pairs, we introduce value-gap weights for positive and negative samples:

$$\beta_{uv} = \frac{\gamma_{\text{pos}} TBC_{\text{median}}}{|\Delta_{uv}|}, \quad \beta_{uw} = \frac{|\Delta_{uw}|}{\gamma_{\text{neg}} TBC_{\text{median}}},$$

which amplify near positives and distant negatives while preventing blow-ups when $|\Delta|$ is small.

Re-weighted contrastive loss. Let $\mathbf{h}_u$ be the backbone embedding (Sec. 4.1), $\text{sim}(\cdot, \cdot)$ the dot-product similarity, and τ the temperature. With $z_{ux} = \beta_{ux} \text{sim}(\mathbf{h}_u, \mathbf{h}_x)/\tau$, the loss is defined as:

$$\mathcal{L}_{\text{contrast}} = \sum_{i=1}^{\hat{k}} \sum_{u \in \psi_i} -\log \frac{\sum_{v \in \mathcal{S}^+(u)} \exp(z_{uv})}{\sum_{v \in \mathcal{S}^+(u)} \exp(z_{uv}) + \sum_{w \in \mathcal{S}^-(u)} \exp(z_{uw})}.$$

This design aligns close-valued nodes tightly while pushing apart far-valued nodes, improving stability and discriminability.

ValueNet. For each node $v \in V$, the TBC score at the largest timestamp t_{max} represents the final TBC value of v under the full-graph setting and is predicted using a three-layer MLP:

$$TBC(v) = TBC_{t_{\text{max}}}(v) = \text{MLP}(h_v^l(t_{\text{max}})).$$

Total Loss Function. A weighted total loss that balances regression and contrastive objectives is defined as:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{contrast}} + (1 - \alpha) \cdot \mathcal{L}_{\text{regress}}.$$

4.3 Discussions

(i) **Design choice of stability-based clustering.** Clustering operates as a fixed pre-processing step rather than being optimized jointly with the contrastive objective. This reflects two considerations: (1) *stability and robustness.* A fully learnable binning strategy easily converges to degenerate solutions (e.g., assigning all low-TBC nodes to a single large cluster), which undermines the learning of meaningful TBC-aware representations and destabilizes training; and (2) *decoupling noisy labeling from representation learning.* By separating pseudo-label generation from GNN optimization, we anchor the contrastive loss to statistically grounded partitions derived from ground-truth TBC values, avoiding the instability of chasing a moving target. Empirically, we compared against adaptive thresholding [22] and learnable binning strategies [42]. Our stability-seeking approach consistently achieves substantially lower MAE and higher Spearman correlation (up to 915% MAE reduction and 88.7% gains in ranking correlation), confirming that stability-seeking clustering provides stronger and more reliable supervision. Due to page limits, detailed results are provided in Appendix D.1.

(ii) **Theoretical guarantees.** The stability-based clustering in KContrastNet (Definition 5) enjoys statistical guarantees, the instability estimator is unbiased and consistent with respect to the bootstrap size, ensuring reliable determination of $\hat{k}$. For contrastive learning, the value-gap weights are monotone in the TBC gap, amplifying informative pairs while preserving smoothness. Furthermore, the re-weighted contrastive loss is bounded, gradients are controlled, and non-collapse is guaranteed. Formal statements and proofs are deferred to Appendix B.

5 Experimental Evaluation

5.1 Experimental Setup

• **Datasets.** We utilize 10 real-world temporal graphs with varying scales and skewness for testing. Table 9 in Appendix C provides their details. All data sets are publicly available from online repositories, including netzscheuler [23], SNAP [18], SocioPatterns [12], Konect [17] and Network Repository [27]. Our training dataset consists of 50 real-world temporal networks from Network Repository [27] and Konect [17]. For every dataset, we repeated each experiment 30 times and report the mean and the standard deviation of all scores. The associated learning rates as well as all other hyperparameters of the models are reported in Appendix C.

• **Compared Baselines.** To evaluate the computational efficiency, we compare CLGNN with the **exact algorithm** ETBC [44]. To evaluate effectiveness, we compare CLGNN against:

(i) **Static GNNs**, including DrBC [10], a GNN-based classifier targeting high BC node identification; and GNN-Bet [20], a supervised model that predicts BC scores from node embeddings.

(ii) **Temporal GNNs**, including TGAT [37], which models time-evolving interactions using attention and continuous time encoding; TATKC [43], which uses time-injected attention for centrality ranking; DBGNN [15], the SOTA transductive temporal GNN framework built upon De Bruijn graphs for centrality prediction; Temporal Transformer [36], which demonstrates the feasibility of the Transformer architecture for dynamic graph modeling; and TempGNTK [33], a kernel-based temporal graph model.

Table 1: MAE on 10 datasets: CLGNN vs. state-of-the-art learning and approximation baselines.

Dataset	CLGNN	DrBC	GNN-Bet	DBGNN	TGAT	TATKC	T-Trans	TempGNTK	IIOIwD	ONBRA
sp_hospital	13.62±0.33	28.86±0.32	31.35±0.26	<u>27.14±0.73</u>	31.07±0.38	31.87±0.45	29.82±0.45	30.83±0.46	31.05±0.19	42.74±0.00
sp_workplace	12.33±0.21	36.68±0.28	36.40±0.53	55.96±0.58	35.52±0.26	36.98±0.50	40.44±0.47	28.72±0.33	35.82±0.31	48.02 ± 0.00
Highschool2011	26.69±0.22	84.48±0.64	75.31±0.48	<u>58.51±0.44</u>	74.70±0.47	75.75±1.00	88.45±0.86	78.66±0.76	74.84±1.22	90.38±0.00
Highschool2012	<u>157.54±1.28</u>	307.25±3.99	193.89±2.23	283.20 ± 9.11	193.43±1.60	194.97±1.39	193.257±2.21	138.82±1.39	193.82±2.50	215.98±0.00
haggle	5.22±0.12	50.17±0.73	92.58±1.15	<u>44.05±0.98</u>	48.85±0.31	48.71±0.31	47.54±0.51	76.77±0.82	48.19±0.88	318.71±0.00
ia-reality-call	170.23±2.90	535.28±9.50	425.77±4.85	<u>218.76±5.81</u>	316.87±3.46	246.85±3.21	1559.17±21.00	1967.70±26.50	297.40±1.89	45604.18±0.00
infectious	106.93±2.05	376.90±5.98	344.98±3.34	<u>122.61±3.21</u>	195.39±1.70	203.15±2.58	412.47±4.10	598.12±6.20	124.77±1.59	—
wikiedits-se	16.54±0.09	211.16±2.99	315.38±3.59	38.85±0.54	138.73±1.52	<u>25.43±0.18</u>	121.84±1.40	82.93±0.95	265.74±2.70	—
superuser	557.28±12.06	14690.53±144.62	9654.33±185.00	<i>OOM</i>	728.01±9.80	<u>606.73±7.53</u>	822.14±10.30	992.67±12.50	1234.66±12.85	—
wikitalk	1.15±0.12	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	—

Notes: “—” indicates that, under the same time budget and constraints as CLGNN, ONBRA failed to find a sufficient number of valid temporal paths; *OOM* indicates the method ran out of memory or exceeded practical resource limits. Best and second-best results are highlighted in **Bold** and underlined, respectively.

Table 2: Spearman correlation on 10 datasets: CLGNN vs. state-of-the-art learning and approximation baselines.

Dataset	CLGNN	DrBC	GNN-Bet	DBGNN	TGAT	TATKC	T-Trans	TempGNTK	IIOIwD	ONBRA
sp_hospital	0.74±0.02	0.53±0.02	-0.55±0.03	0.069±0.01	-0.15±0.01	-0.05±0.02	0.6750±0.02	0.4165±0.02	-0.29±0.03	0.4871±0.00
sp_workplace	<u>0.45±0.02</u>	0.04±0.02	-0.35±0.01	0.13±0.03	-0.50±0.02	-0.19±0.03	-0.2229±0.03	0.3642±0.03	-0.04±0.01	0.61 ± 0.02
Highschool2011	<u>0.52±0.03</u>	0.46±0.02	-0.39±0.02	0.25±0.02	0.17±0.03	0.28±0.03	0.1100±0.03	0.1636±0.03	-0.38±0.03	0.5405±0.00
Highschool2012	<u>0.51±0.02</u>	0.41±0.03	-0.21±0.01	0.14±0.01	0.23±0.01	0.23±0.03	0.182±0.03	-0.1632±0.03	-0.43±0.03	0.5289±0.02
haggle	0.99±0.03	0.44±0.02	0.06±0.03	-0.884±0.02	0.42±0.03	0.47±0.02	0.6040±0.03	-0.0931±0.03	-0.57±0.02	<u>0.7021±0.03</u>
ia-reality-call	<u>0.47±0.01</u>	0.14±0.03	-0.24±0.03	0.50±0.02	0.18±0.02	0.11±0.03	0.1690±0.02	-0.0202±0.02	-0.48±0.03	0.1707±0.00
infectious	0.55±0.02	0.30±0.02	-0.10±0.01	0.20±0.01	0.22±0.02	0.41±0.01	0.235±0.02	0.052±0.02	-0.49±0.03	—
wikiedits-se	0.51±0.02	0.28±0.03	-0.22±0.02	0.26±0.03	0.19±0.03	0.23±0.03	0.218±0.03	0.041±0.02	-0.25±0.02	—
superuser	0.41±0.03	0.12±0.03	-0.39±0.01	<i>OOM</i>	0.16±0.02	0.20±0.03	0.182±0.03	0.071±0.02	-0.54±0.03	—
wikitalk	0.19±0.02	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	<i>OOM</i>	—

Table 3: Runtime (s) of CLGNN vs. ETBC across datasets.

Methods	sp_hospital	sp_workplace	Highschool2011	Highschool2012	haggle	ia-reality-call	infectious	wikiedits-se	superuser	wikitalk
CLGNN (s)	0.94	0.15	1.00	1.49	0.83	1.14	1.95	1.81	20.22	179.90
ETBC (s)	14.15	1.16	4.81	30.54	16.53	257.84	1029.83	1201.33	5388.36	31100.11

Table 4: MAE of CLGNN vs. DBGNN across discretized TBC ranges. $\Delta(\%) = \frac{\text{CLGNN}-\text{DBGNN}}{\text{DBGNN}} \times 100\%$.

Dataset	zero-value			mid-value			high-value		
	CLGNN	DBGNN	$\Delta(\%)$	CLGNN	DBGNN	$\Delta(\%)$	CLGNN	DBGNN	$\Delta(\%)$
sp_hospital	2.04±0.05	10.07±0.27	-79.74%	4.77±0.12	8.83±0.24	-45.98%	6.81±0.17	8.24±0.22	-17.35%
sp_workplace	1.12±0.02	18.05±0.19	-93.80%	4.48±0.08	20.51±0.21	-78.16%	6.73±0.11	17.40±0.18	-61.32%
Highschool2011	6.35±0.05	17.90±0.13	-64.53%	7.63±0.06	18.71±0.14	-59.22%	12.71±0.10	21.90±0.16	-41.96%
Highschool2012	18.18±0.15	94.05±3.03	-80.67%	54.53±0.44	96.09±3.09	-43.25%	84.83±0.69	93.06±2.99	-8.84%
haggle	0.67±0.02	22.22±0.49	-96.98%	1.40±0.03	12.46±0.28	-88.76%	3.15±0.07	9.37±0.21	-66.38%
ia-reality-call	0.0025±0.00	0.23±0.01	-98.91%	4.21±0.00	45.88±1.22	-90.82%	166.02±0.00	172.65±4.59	-3.84%
infectious	15.28±0.29	26.41±0.69	-42.14%	38.21±0.73	46.53±1.22	-17.88%	53.44±1.03	49.67±1.30	+7.59%
wikiedits-se	2.08±0.01	0.79±0.01	+163.29%	7.69±0.04	10.70±0.15	-28.13%	6.77±0.04	27.36±0.38	-75.26%
superuser	43.14±0.93	<i>OOM</i>	—	183.36±3.97	<i>OOM</i>	—	330.77±7.16	<i>OOM</i>	—
wikitalk	0.15±0.02	<i>OOM</i>	—	0.40±0.04	<i>OOM</i>	—	0.60±0.06	<i>OOM</i>	—

(iii) **Contrastive learning models**, IIOIwD [45], which estimates node importance using contrastive sampling.

(iv) **Approximation algorithm**, ONBRA [29], a rigorous estimator of temporal betweenness centrality in temporal networks. For fairness, we evaluate ONBRA under the same wall-clock inference time as CLGNN on each dataset.

• **Evaluation Metrics.** We mainly use two metrics to assess performance: mean absolute error (MAE) for prediction accuracy, Spearman correlation for ranking quality.

• **Reproducibility.** Code of CLGNN is available at <https://github.com/EmoryChang/CLGNN/>.

Next, we seek to answer the following research questions:

RQ1 (Effectiveness, Scalability & Robustness). How does CLGNN perform in terms of predictive accuracy, scalability to massive temporal graphs, and robustness across datasets with varying skewness, compared to representative baselines?

RQ2 (Efficiency). How much faster is CLGNN compared with the leading exact method ETBC?

RQ3 (Stratified Performance). How well does CLGNN perform across discretized TBC ranges (zero, mid, high) relative to the state-of-the-art TBC prediction methods?

RQ4 (Ablation: Necessity of Components). Are contrastive learning, path-time dual-aware encoding, and stability-based clustering each essential for CLGNN’s performance?

RQ5 (Imbalance Learning). How does CLGNN’s stability-guided contrastive learning framework compare to alternative imbalance-handling strategies?

RQ6 (Parameter Sensitivity). How sensitive is CLGNN’s performance to changes in key hyperparameters?

RQ7 (Generalizability). Can CLGNN generalize beyond temporal betweenness centrality to other temporal centrality measures or alternative optimal-path definitions?

Table 5: Ablation of CLGNN components: $\Delta(\%) = \frac{\text{Variant-CLGNN}}{\text{CLGNN}} \times 100\%$.

Dataset	w/o Contrastive Learning				w/o Path-Time Encoder				w/o Stability-based Clustering			
	MAE	Spearman	$\Delta\text{MAE}(\%)$	$\Delta\text{Spearman}(\%)$	MAE	Spearman	$\Delta\text{MAE}(\%)$	$\Delta\text{Spearman}(\%)$	MAE	Spearman	$\Delta\text{MAE}(\%)$	$\Delta\text{Spearman}(\%)$
sp_hospital	30.82±0.51	0.280±0.03	126.28%	-62.01%	54.48±1.63	0.405±0.03	300.00%	-45.05%	49.03±1.72	0.074±0.04	259.99%	-89.96%
sp_workplace	36.38±0.24	-0.120±0.03	141.73%	-135.29%	39.50±1.19	0.150±0.03	162.46%	-55.88%	36.13±1.26	0.020±0.04	140.07%	-94.12%
Highschool2011	75.27±0.69	-0.360±0.01	182.02%	-169.23%	73.42±2.20	0.380±0.03	175.20%	-26.92%	79.96±2.80	0.020±0.04	199.56%	-96.15%
Highschool2012	193.86±3.28	-0.26±0.00	23.05%	-150.98%	192.23±1.23	0.200±0.01	22.03%	-60.78%	210.71±3.10	0.14±0.01	33.76%	-72.55%
haggle	52.25±0.62	0.490±0.01	900.96%	-50.51%	48.83±1.46	0.290±0.03	835.06%	-70.71%	39.83±1.39	0.040±0.04	662.26%	-95.96%
ia-reality-call	209.38±1.85	0.330±0.02	22.998%	-29.79%	374.42±11.23	0.211±0.03	120.01%	-55.11%	476.53±16.68	0.070±0.04	180.01%	-85.11%
infectious	148.20±3.10	0.38±0.02	38.60%	-30.91%	228.74±6.90	0.27±0.03	113.92%	-50.91%	302.15±9.80	0.12±0.03	182.57%	-78.18%
wikiedits-se	25.92±0.40	0.33±0.03	56.71%	-35.29%	34.10±0.85	0.24±0.03	106.17%	-52.94%	41.85±1.10	0.08±0.04	153.02%	-84.31%
superuser	691.24±5.04	0.160±0.03	24.04%	-61.35%	1,058.83±31.77	0.207±0.03	89.95%	-50.00%	1,448.93±50.71	0.062±0.04	160.00%	-85.02%
wikitalk	2.10±0.08	0.07±0.03	82.61%	-63.16%	2.95±0.12	0.05±0.03	156.52%	-73.68%	3.60±0.15	-0.02±0.04	213.04%	-110.53%

Table 6: Ablation (Imbalance Learning): $\Delta(\%) = \frac{\text{Variant-CLGNN}}{\text{CLGNN}} \times 100\%$.

Dataset	Over- & Under- sampling				Tweedie Loss (Weighted)				Graph Aug (Random)			
	MAE	Spearman	$\Delta\text{MAE}(\%)$	$\Delta\text{Spearman}(\%)$	MAE	Spearman	$\Delta\text{MAE}(\%)$	$\Delta\text{Spearman}(\%)$	MAE	Spearman	$\Delta\text{MAE}(\%)$	$\Delta\text{Spearman}(\%)$
sp_hospital	149.01±2.78	-0.05±0.02	+994.05%	-106.78%	30.85±0.46	-0.07±0.03	+126.51%	-109.50%	34.93±0.50	0.22±0.03	+156.46%	-70.15%
sp_workplace	65.67±0.70	-0.45±0.02	+336.35%	-232.35%	32.40±0.22	0.13±0.01	+115.28%	-61.76%	29.00±0.58	-0.16±0.02	+92.69%	-147.06%
Highschool2011	505.77±4.94	0.20±0.01	+1794.98%	-61.54%	69.49±0.43	-0.13±0.02	+160.36%	-125.00%	89.18±1.02	-0.04±0.03	+234.13%	-107.69%
Highschool2012	1,428.01±14.72	0.12±0.03	+806.45%	-76.47%	184.34±1.98	-0.41±0.01	+17.02%	-180.39%	181.98±3.43	-0.15±0.01	+15.52%	-129.41%
haggle	1,670.79±18.09	0.58±0.02	+31907.47%	-41.41%	169.75±1.99	-0.03±0.03	+3151.92%	-103.03%	258.64±3.30	0.54±0.03	+4854.79%	-45.45%
ia-reality-call	1,022.29±18.85	0.28±0.03	+500.68%	-40.30%	234.29±2.42	0.23±0.02	+37.66%	-50.96%	293.25±4.73	0.29±0.01	+72.31%	-38.17%
infectious	1,668.81±11.99	0.33±0.02	+1460.66%	-40.00%	194.25±2.23	0.08±0.02	+81.66%	-85.45%	212.49±1.34	-0.03±0.03	+98.72%	-105.45%
wikiedits-se	584.82±7.23	0.35±0.03	+3435.79%	-31.37%	263.23±4.48	0.08±0.01	+1491.48%	-84.31%	327.70±2.64	0.17±0.03	+1881.26%	-66.67%
superuser	1,660.90±28.27	0.54±0.03	+198.04%	+30.43%	746.62±4.27	0.20±0.02	+33.98%	-51.69%	927.87±4.74	0.02±0.03	+66.50%	-95.17%
wikitalk	6.80±0.25	0.12±0.03	+491.30%	-36.84%	1.45±0.06	0.16±0.02	+26.09%	-15.79%	1.95±0.07	0.10±0.03	+69.57%	-47.37%

Table 7: Comparison of CLGNN vs. DBGNN under alternative centrality metrics (temporal closeness and Katz centralities).

Dataset	temporal closeness centralities				temporal Katz centralities			
	CLGNN		DBGNN		CLGNN		DBGNN	
	MAE	Spearman	MAE	Spearman	MAE	Spearman	MAE	Spearman
sp_hospital	6.03 ± 0.09	0.70 ± 0.02	24.91 ± 0.41	0.22 ± 0.02	6.48 ± 0.11	0.70 ± 0.02	20.74 ± 0.34	0.20 ± 0.02
sp_workplace	13.44 ± 0.17	0.51 ± 0.01	41.32 ± 0.62	0.19 ± 0.02	10.41 ± 0.16	0.47 ± 0.02	33.88 ± 0.59	0.17 ± 0.02
Highschool2011	2.73 ± 0.03	0.552 ± 0.027	3.36 ± 0.04	0.206 ± 0.019	2.453849 ± 0.021	0.1903 ± 0.020	2.974736 ± 0.031	0.187185 ± 0.020
Highschool2012	10.62 ± 0.15	0.461 ± 0.016	29.10 ± 0.35	0.128 ± 0.02	12.28 ± 0.21	0.486 ± 0.018	36.92 ± 0.61	0.151 ± 0.02
haggle	1.94 ± 0.03	0.94 ± 0.02	60.98 ± 0.63	0.92 ± 0.02	1.09 ± 0.0250	0.97 ± 0.010	3.86 ± 0.04	0.94 ± 0.01
ia-reality-call	20.35 ± 0.29	0.51 ± 0.02	30.22 ± 0.38	0.53 ± 0.02	18.66 ± 0.25	0.49 ± 0.01	27.04 ± 0.35	0.51 ± 0.02
infectious	2.11 ± 0.04	0.48 ± 0.02	42.51 ± 0.44	0.26 ± 0.02	1.37 ± 0.02	0.26 ± 0.02	2.21 ± 0.03	0.25 ± 0.02
wikiedits-se	10.02 ± 0.06	0.53 ± 0.01	26.45 ± 0.28	0.27 ± 0.02	13.27 ± 0.10	0.52 ± 0.019	34.72 ± 0.41	0.25 ± 0.03
superuser	41.12 ± 1.15	0.45 ± 0.03	OOM	OOM	37.66 ± 1.06	0.44 ± 0.03	OOM	OOM
wikitalk	1.71 ± 0.05	0.14 ± 0.01	OOM	OOM	1.16 ± 0.10	0.18 ± 0.02	OOM	OOM

Table 8: Comparison of CLGNN vs. DBGNN across alternative optimal temporal path definitions.

Dataset	Shortest				Shortest + Earliest Arrival				Shortest + Latest Departure + Earliest Arrival			
	CLGNN		DBGNN		CLGNN		DBGNN		CLGNN		DBGNN	
	MAE	Spearman	MAE	Spearman	MAE	Spearman	MAE	Spearman	MAE	Spearman	MAE	Spearman
sp_hospital	7.94 ± 0.11	0.7213 ± 0.01	15.83 ± 0.20	0.07 ± 0.02	13.62 ± 0.33	0.74 ± 0.02	27.14 ± 0.73	0.07 ± 0.01	25.45 ± 0.2458	0.51 ± 0.012	50.72 ± 0.51	0.05 ± 0.02
sp_workplace	16.72 ± 0.18	0.54 ± 0.01	75.87 ± 0.76	0.16 ± 0.02	12.33 ± 0.21	0.45 ± 0.02	55.96 ± 0.58	0.13 ± 0.03	36.18 ± 0.16	0.36 ± 0.01	164.20 ± 1.64	0.10 ± 0.02
Highschool2011	33.44 ± 0.19	0.56 ± 0.03	43.25 ± 0.43	0.21 ± 0.02	26.69 ± 0.22	0.52 ± 0.03	58.51 ± 0.44	0.25 ± 0.02	37.36 ± 0.40	0.40 ± 0.03	93.21 ± 0.93	0.16 ± 0.02
Highschool2012	123.86 ± 0.71	0.4763 ± 0.02	1640.72 ± 16.41	-0.1628 ± 0.02	157.54 ± 1.28	0.51 ± 0.02	283.20 ± 9.11	0.14 ± 0.01	197.98 ± 1.16	0.39 ± 0.0254	1621.85 ± 16.22	-0.1521 ± 0.02
haggle	5.00 ± 0.07	0.99 ± 0.02	42.15 ± 0.42	-0.884 ± 0.02	5.22 ± 0.12	0.99 ± 0.03	44.05 ± 0.98	-0.884 ± 0.02	6.46 ± 0.09	0.73 ± 0.01	54.55 ± 0.55	-0.65 ± 0.02
ia-reality-call	149.00 ± 1.58	0.51 ± 0.02	191.48 ± 1.91	0.54 ± 0.02	170.23 ± 2.90	0.47 ± 0.01	218.76 ± 5.81	0.50 ± 0.02	186.90 ± 2.73	0.30 ± 0.03	240.18 ± 2.40	0.32 ± 0.02
infectious	89.25 ± 1.05	0.59 ± 0.02	102.34 ± 1.02	0.21 ± 0.02	106.93 ± 2.05	0.55 ± 0.02	122.61 ± 3.21	0.20 ± 0.01	100.57 ± 1.30	0.39 ± 0.02	115.31 ± 1.15	0.14 ± 0.02
wikiedits-se	10.67 ± 0.0451	0.5313 ± 0.014	25.06 ± 0.25	0.27 ± 0.02	16.54 ± 0.09	0.51 ± 0.02	38.85 ± 0.54	0.26 ± 0.03	25.89 ± 0.19	0.41 ± 0.02	60.81 ± 0.61	0.21 ± 0.02
superuser	534.78 ± 7.04	0.44 ± 0.03	OOM	OOM	557.28 ± 12.06	0.41 ± 0.03	OOM	OOM	592.96 ± 6.83	0.25 ± 0.01	OOM	OOM
wikitalk	1.85 ± 0.03	0.14 ± 0.01	OOM	OOM	1.15 ± 0.12	0.19 ± 0.02	OOM	OOM	3.40 ± 0.06	0.06 ± 0.01	OOM	OOM

5.2 Experimental Results.

RQ1 (Effectiveness, Scalability & Robustness). As shown in Tables 1 and 2, we observe that (i) CLGNN consistently outperforms static GNN baselines (DrBC and GNN-Bet) across all datasets and metrics, achieving $1.23 \times$ to $26.36 \times$ lower MAE and up to $16.50 \times$ higher Spearman correlation. The gains are especially prominent on datasets with high temporal variation (e.g., *sp_hospital*, *haggle*, *ia-reality-call*); (ii) Against temporal GNNs (DBGNN, TGAT, TATKC, T-Trans, TempGNTK), CLGNN attains the lowest MAE and yields the highest Spearman correlation on 9 out of 10 datasets, matching DBGNN on *ia-reality-call*. Particularly, CLGNN reduces MAE by up to $8.44 \times$ and improves Spearman correlation by up to $3.64 \times$ on datasets like *Highschool2011* and *ia-reality-call*, highlighting its

robustness in dynamic settings. DBGNN performs well on some datasets because it is a transductive model that splits a single graph into training and testing sets, potentially leveraging more direct structural overlap; (iii) Regarding the approximation algorithm ONBRA, CLGNN achieves $1.37 \times$ to $267.90 \times$ lower MAE. ONBRA fails to find valid paths on medium- and large-scale datasets under the same running time; (iv) In terms of the contrastive learning model IIOIWd, CLGNN consistently achieves lower errors and stronger ranking performance, with $1.1 \times$ to $16.07 \times$ smaller MAE, and improves Spearman by $+0.49$ to $+1.56$. This confirms that CLGNN’s temporal path encoding and stability-guided contrastive learning are more effective than IIOIWd’s sampling-based contrastive design. Finally, CLGNN demonstrates stable performance across datasets, scales effectively to large graphs such as *wikitalk*, and maintains

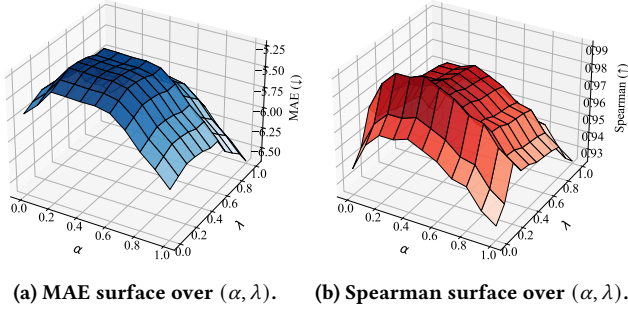


Figure 3: Hyperparameter tuning on *hagggle*: joint performance surfaces for (a) MAE and (b) Spearman.

superiority across networks with diverse structures and skewness levels, highlighting both its scalability and robustness.

RQ2 (Efficiency). We compare the efficiency of CLGNN with the exact TBC algorithm. As shown in Table 3, CLGNN achieves substantial speedup, ranging from 4.8 $\times$ to 663.7 $\times$, across all datasets.

RQ3 (Stratified Performance). CLGNN shows significant improvements across all TBC ranges, as shown in Table 4. It reduces MAE by up to 98.91% for zero-value nodes, 90.82% for mid-value nodes, and 75.26% lower MAE for high-value nodes. Notably, on highly skewed datasets (e.g., *ia-reality-call*: skewness 18.39; *wikiedits-se*: skewness 52.85), CLGNN achieves particularly large gains for mid- and high-value nodes.

RQ4 (Ablation: Necessity of Components). Table 5 shows that removing any core module significantly degrades performance. Without contrastive learning, MAE increases by 23% to 901%, while Spearman drops by 30% to 169%, even turning negative on some datasets. Omitting the path-time encoder raises MAE by 22% to 835% and reduces Spearman by 27% to 74%. Removing stability-based clustering increases MAE by 34% to 662% and decreases Spearman by 73% to 111%. These results confirm that each component is necessary. Contrastive learning stabilizes ranking under skew by shaping hard negatives; the path-time encoder captures indispensable temporal and path cues; stability-based clustering mitigates head-tail imbalance by structuring contrastive pairs.

RQ5 (Imbalance Learning). Table 6 presents the impact of different imbalance-handling techniques on CLGNN. We observe that, (i) Over- & Under-sampling drastically worsens performance, with MAE increasing by 200% to 31,900% and Spearman correlation dropping by 30% to 230%. This indicates that naive resampling can severely distort temporal and structural dependencies. (ii) Tweedie Loss (Weighted) moderately degrades performance, with MAE rising by 17% to 3,152% and Spearman decreasing by 16% to 180%. While weighting helps account for extreme nodes, it is insufficient alone to fully correct for extreme distribution imbalances. (iii) Random Graph Augmentation also negatively affects results, leading to MAE increases of +16% to +4,855% and Spearman drops of 38% to 147%, suggesting that indiscriminate augmentation can disrupt essential graph topology.

RQ6 (Parameter Sensitivity). Figure 3 evaluates the sensitivity of CLGNN to the loss trade-off coefficient α and the aggregation weight λ . The best results are obtained when α is set to 0.2 and λ falls

within 0.2 ~ 0.4. This demonstrates that CLGNN’s performance is sensitive to these hyperparameters, and careful tuning is necessary to achieve an optimal balance between regression accuracy and representation quality.

RQ7 (Generalizability). All the above experimental results are obtained based on the commonly used shortest and earliest arrival temporal paths, which minimize hops and arrival time. However, CLGNN generalizes to alternative temporal path definitions and other temporal centrality measures.

From Table 7, CLGNN consistently outperforms DBGNN on both temporal closeness and temporal Katz centralities, achieving the lowest MAE on all datasets and the highest Spearman on 9 out of 10 datasets (with comparable performance on *ia-reality-call*). For temporal closeness, CLGNN reduces MAE by 1.23 $\times$ to 31.43 $\times$ and increases Spearman by 2% to 260%. For temporal Katz, it reduces MAE by 1.21 $\times$ to 3.54 $\times$ and boosts Spearman by up to 250%. Notably, CLGNN scales to large graphs where DBGNN runs out of memory (e.g., *superuser*, *wikitalk*).

From Table 8, CLGNN also adapts to different path rules. Across shortest-only, shortest + earliest arrival, and shortest + latest departure + earliest arrival settings, CLGNN achieves MAE reductions of 1.15 $\times$ to 13.25 $\times$ and Spearman gains of 95% to 957% relative to DBGNN. Importantly, Spearman for CLGNN remains positive where DBGNN is weak or negative. While both methods degrade as path constraints tighten, CLGNN deteriorates more slowly. The decline reflects the increasing difficulty of modeling both forward and backward temporal dependencies under tighter path rules.

Overall, these results demonstrate that CLGNN generalizes effectively. It is not restricted to TBC but extends to other temporal centralities and diverse path definitions, consistently maintaining accuracy, rank stability, and scalability. Additionally, we evaluate HitsIn@K scores (HitsIn10, 30, 50) to assess how well the top predictions match the ground truth. Due to space constraints, the results are presented in Tables 11, 12, and 13 in Appendix D.2. We observe that (i) CLGNN achieves top or near-top HitsIn@K across multiple datasets and thresholds; (ii) CLGNN maintains performance on large-scale datasets where several baselines fail; and (iii) the relative advantage of CLGNN increases with K, indicating that its performance gap widens as the top-K threshold grows.

6 Conclusion

This paper proposes CLGNN, a contrastive learning-based graph neural network designed to accurately predict TBC. CLGNN incorporates a dual aggregation mechanism that combines temporal path count encoding and time encoding to learn expressive node representations. It also includes a stability-based clustering-guided contrastive learning module to better distinguish nodes across different TBC value ranges. Extensive experiments on real-world datasets with varying sizes and skewness demonstrate the effectiveness, scalability, and generalizability of CLGNN. For future work, we plan to explore advanced temporal graph augmentation techniques to further improve model generalization and robustness.

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A Algorithm and Complexity Analysis

A.1 Stability-Based Clustering: Algorithm and Complexity

Time complexity. We analyze the time complexity of Algorithm 1, which estimates the optimal number of clusters k using a stability-based bootstrap method.

Notation. Let N denote the total number of feature samples in the input set F , and let $r \in (0, 1)$ be the stratified sampling ratio. Let $n = rN$ be the size of the reduced sample set F^n . Let B be the number of bootstrap resample pairs, and K the maximum number of candidate clusters.

Step 1: Sampling and Bootstrapping.

- Stratified sampling requires $O(N)$ time.
- Constructing B independent resample pairs of size n requires $O(Bn)$.

Step 2: Clustering Instability Computation. For each candidate cluster number $k \in \{2, \dots, K\}$, and each bootstrap pair (X_i, Y_i) , the following operations are performed:

Algorithm 1: Stability-Based Estimation of Optimal Cluster Number $\hat{k}$

```

1 Inputs: Feature set  $F=\{f_1, \dots, f_N\}$ , sampling rate  $r \in (0, 1)$ ,
   #bootstrap pairs  $B$ , candidates  $k \in \{2, \dots, K\}$ ;
2 Output:  $\hat{k}$ ;
3 Step 1: Sampling and Bootstrap
4  $F^n \leftarrow \text{StratifiedSample}(F, r)$ ;
    $\{(X_i, Y_i)\}_{i=1}^B \leftarrow \text{BootstrapResamplePair}(F^n)$ ;
5 Step 2: Compute clustering instability for each  $k$ 
6 for  $k=2$  to  $K$  do
7    $s_k \leftarrow 0$ ;
8   for  $i=1$  to  $B$  do
9      $\psi_{X_i} \leftarrow \text{KMeans}(X_i, k)$ ;  $\psi_{Y_i} \leftarrow \text{KMeans}(Y_i, k)$ ;
10     $s_k \leftarrow s_k + d(\psi_{X_i}, \psi_{Y_i})$ ; // Def. 4
11   $\hat{s}_B(k) \leftarrow s_k/B$ ; // instability for  $k$ 
12 return  $\hat{k} \leftarrow \arg \min_{k \in \{2, \dots, K\}} \hat{s}_B(k)$ ;

```

- **K-Means clustering:** Two independent K-Means clusterings of size n and k clusters take $O(nkT)$ time each, where T is the number of iterations (assumed constant in practice).
- **Clustering distance computation:** For each pair of clusterings, computing the pairwise agreement distance (as defined in Definition 4) over $O(n^2)$ pairs requires $O(n^2)$ time.

Therefore, the total time complexity for instability estimation is:

$$O\left(\sum_{k=2}^K \sum_{i=1}^B (2nkT + n^2)\right) = O\left(B \cdot \sum_{k=2}^K (2nkT + n^2)\right) = O(BnTK^2 + Bn^2K)$$

Remarks. This complexity is dominated by the clustering instability computations in Step 2. In practice, setting $B \ll N$, $r \ll 1$, and bounding K (e.g., $K < 10$) ensures tractability. Further efficiency can be achieved by approximating the clustering distance or subsampling intra-pair comparisons.

Wall-clock example. On the training set *Hyccups*, the **KContrastNet-Clustering** and **KContrastNet-Contrastive Learning** components take 2.807s (47.2%) and 0.072s (1.21%) of the per-epoch training time, respectively. These costs are incurred only during training and are removed at inference.

Space complexity. For the main memory cost, storing the sub-sampled embeddings requires $O(n \cdot d)$ space, where d is the embedding dimension. Pairwise distance computation in the stability-based clustering incurs a cost of $O(n^2)$. Therefore, the overall space complexity of the module is $O(n^2 + n \cdot d)$.

A.2 CLGNN: Overall Complexity

We summarize the training complexity and the inference-time complexity of CLGNN. The per-epoch training cost of the *representation module* is $O(LH M_{\text{eff}} d_{\text{msg}} d_h)$, arising from backbone message passing. Here, $M_{\text{eff}} = |\{(u, v, t) \in E : P(u, v, t) > 0\}|$ is the number of

effective messages; L is the number of layers, H the number of attention heads, and d_{msg}, d_h are the message and hidden dimensions. The *prediction module* incurs additional cost mainly from bootstrap clustering, $O(BnTK^2 + Bn^2K)$ and contrastive pair construction, $O(P_{\text{ctr}} d_h)$. Here, $P_{\text{ctr}} = \sum_{u \in V} (|S^+(u)| + |S^-(u)|)$, B is the number of bootstrap pairs, T the number of k -means iterations, and n is the size of the sampled subset. Thus, the overall per-epoch training cost of CLGNN is $O(LH M_{\text{eff}} d_{\text{msg}} d_h + BnTK^2 + Bn^2K + P_{\text{ctr}} d_h)$. Notably, the additional terms from bootstrap clustering and contrastive pair construction are training-only (computed on a sampled subset with bounded pair budgets) and are removed at inference, so the runtime matches a standard GNN forward pass. At inference, clustering and pairwise similarity computations are removed. Only the backbone forward pass and the ValueNet are executed, resulting in a runtime $O(LH M_{\text{eff}} d_{\text{msg}} d_h)$, the same as a standard GNN forward.

B Theoretical Properties

B.1 Unbiasedness & Consistency of Instability Estimator

Let $Z_i := d(\psi_{X_i}, \psi_{Y_i})$, which is a real-valued random variable. The instability estimator for cluster number k defined in Definition 5 is

$$\hat{s}_B(\psi, k, n) = \frac{1}{B} \sum_{i=1}^B Z_i.$$

Unbiasedness. Because $\{Z_i\}_{i=1}^B$ are i.i.d. realizations of Z , the empirical mean satisfies

$$\mathbb{E}[\hat{s}_B(\psi, k, n)] = \frac{1}{B} \sum_{i=1}^B \mathbb{E}[Z_i] = \mathbb{E}[Z].$$

Thus $\hat{s}_B(\psi, k, n)$ is an unbiased estimator of the expected clustering instability.

Consistency. By the Weak Law of Large Numbers (WLLN), for any $\epsilon > 0$,

$$\Pr(|\hat{s}_B(k) - \mathbb{E}[Z]| > \epsilon) \rightarrow 0, \quad \text{as } B \rightarrow \infty.$$

Hence, $\hat{s}_B(k) \xrightarrow{P} \mathbb{E}[d(\psi_X, \psi_Y)]$, i.e., the estimator is consistent.

B.2 Monotonicity of Value-gap Weights

For $\beta_{uv} = \frac{TBC_{\text{median}} \cdot \gamma_{\text{pos}}}{\Delta_{uv}}$, we have:

$$\frac{d\beta_{uv}}{d\Delta_{uv}} = -\frac{TBC_{\text{median}} \cdot \gamma_{\text{pos}}}{\Delta_{uv}^2} < 0.$$

Hence, β_{uv} decreases as the TBC difference increases, emphasizing the similarity of closely matched pairs.

For $\beta_{uw} = \frac{\Delta_{uw}}{TBC_{\text{median}} \cdot \gamma_{\text{neg}}}$, we have:

$$\frac{d\beta_{uw}}{d\Delta_{uw}} = \frac{1}{TBC_{\text{median}} \cdot \gamma_{\text{neg}}} > 0.$$

Therefore, β_{uw} increases linearly with TBC difference, encouraging separation of semantically distant pairs.

Table 9: Datasets for evaluation. $|V|$: the number of vertices, $|E|$: the number of edges, $|T|$: the number of distinct timestamps. We also report TBC skewness and three topology states: clustering coefficient (C), and density. “—” indicates the statistic could not be computed due to dataset scale/resource constraints.

Dataset	$ V $	$ E $	$ T $	Skewness	C	Density	Description
sp_hospital [35]	75	32,424	9,453	2.26	0.640280	0.410450	Face-to-face interactions in a hospital
sp_workplace [13]	92	755	9,827	2.64	0.213000	0.090200	Face-to-face interactions in a workspace
Highschool2011 [12]	126	28,561	5,609	2.19	0.575908	0.217016	High school social network 2011
Highschool2012 [12]	180	45,047	11,273	2.53	0.475204	0.137803	High school social network 2012
haggle [5]	274	28,244	15,662	4.06	0.632680	0.056790	Human proximity recorded by smart devices
ia-reality-call [8]	6,809	52,050	50,781	18.39	0.017771	0.000331	Reality-based call dataset
infectious [16]	10,972	415,912	76,944	1.54	0.446670	0.000740	Infectious disease contact data
wikiedits-se [17]	18,055	261,169	258,625	52.85	0.081860	0.000936	Wikipedia edits (SE) dataset
superuser [18]	194,085	1,443,339	1,437,199	97.49	0.117174	0.000039	Comments, questions, and answers on Super User
wikitalk [18]	1,140,149	7,833,140	7,375,042	777.72	—	0.0000554	Users and discussions from Wikipedia inception

Table 10: Comparison between adaptive thresholding and learnable quantile-based binning on discretization sensitivity.

$$\Delta(\%) = \frac{\text{Variant}-\text{CLGNN}}{\text{CLGNN}} \times 100\%.$$

Dataset	Adaptive thresholding (variance-based)				Learnable quantile-based binning			
	MAE	Spearman	$\Delta\text{MAE}(\%)$	$\Delta\text{Spearman}(\%)$	MAE	Spearman	$\Delta\text{MAE}(\%)$	$\Delta\text{Spearman}(\%)$
sp_hospital	60.10 $\pm$ 1.20	0.35 $\pm$ 0.02	+341.3%	−52.7%	62.50 $\pm$ 1.25	0.22 $\pm$ 0.02	+358.9%	−70.3%
sp_workplace	46.43 $\pm$ 0.90	0.24 $\pm$ 0.02	+276.5%	−46.7%	50.42 $\pm$ 1.00	0.23 $\pm$ 0.02	+308.9%	−48.9%
Highschool2011	83.91 $\pm$ 1.68	0.38 $\pm$ 0.02	+214.5%	−26.9%	84.58 $\pm$ 1.70	0.22 $\pm$ 0.02	+216.8%	−57.7%
Highschool2012	195.97 $\pm$ 3.00	0.17 $\pm$ 0.01	+24.4%	−66.7%	203.52 $\pm$ 3.10	0.10 $\pm$ 0.01	+29.2%	−80.4%
Highschool2013	284.53 $\pm$ 4.00	0.06 $\pm$ 0.01	+77.2%	−88.7%	233.20 $\pm$ 3.50	0.08 $\pm$ 0.01	+45.2%	−84.9%
haggle	52.99 $\pm$ 0.80	0.41 $\pm$ 0.02	+915.0%	−58.6%	44.39 $\pm$ 0.70	0.20 $\pm$ 0.02	+750.0%	−79.8%
ia-reality-call	310.00 $\pm$ 4.65	0.28 $\pm$ 0.02	+82.1%	−40.4%	325.00 $\pm$ 4.90	0.18 $\pm$ 0.02	+90.9%	−61.7%
infectious	160.00 $\pm$ 2.40	0.34 $\pm$ 0.02	+49.6%	−38.2%	168.00 $\pm$ 2.60	0.26 $\pm$ 0.02	+57.1%	−52.7%
wikiedits-se	29.00 $\pm$ 0.45	0.28 $\pm$ 0.02	+75.3%	−45.1%	31.50 $\pm$ 0.50	0.24 $\pm$ 0.02	+90.4%	−52.9%
superuser	980.00 $\pm$ 19.60	0.28 $\pm$ 0.02	+75.8%	−31.7%	1020.00 $\pm$ 20.40	0.20 $\pm$ 0.02	+83.1%	−51.2%
wikitalk	2.30 $\pm$ 0.05	0.10 $\pm$ 0.01	+100.0%	−47.4%	1.80 $\pm$ 0.04	0.12 $\pm$ 0.01	+56.5%	−36.8%

B.3 Boundedness, Gradient Control, and Non-collapse of Contrastive Loss

Boundedness. Each term in the loss defined in Section 4 has the form:

$$\ell_u = -\log\left(\frac{A}{A+B}\right) = \log\left(1 + \frac{B}{A}\right),$$

where $A = \sum_{v \in S^+(u)} \exp(z_{uv}) > 0$, and $B = \sum_{w \in S^-(u)} \exp(z_{uw}) \geq 0$. Since $\text{sim}(\mathbf{h}_u, \mathbf{h}_x) \in [-1, 1]$ and $\beta \leq \beta_{\max}$, we have $B \leq |S^-(u)|e^{\beta_{\max}/\tau}$ and $A \geq |S^+(u)|e^{-\beta_{\max}/\tau}$, therefore,

$$0 \leq \ell_u \leq \log\left(1 + \frac{|S^-(u)|}{|S^+(u)|} e^{2\beta_{\max}/\tau}\right).$$

Thus $\mathcal{L}_{\text{contrast}} = \sum_u \ell_u$ is bounded.

Gradient Bound & Non-collapse. Let $p_{ux} = \frac{e^{z_{ux}}}{\sum_y e^{z_{uy}}}$. Then the gradient of the per-sample loss ℓ_u satisfies

$$\nabla_{\mathbf{h}_u} \ell_u = \frac{1}{\tau} \left[\sum_{v \in S^+(u)} \beta_{uv} p_{uv} (\mathbf{h}_u - \mathbf{h}_v) - \sum_{w \in S^-(u)} \beta_{uw} p_{uw} (\mathbf{h}_u - \mathbf{h}_w) \right].$$

With ℓ_2 -normalized embeddings, we have the bound

$$\|\nabla_{\mathbf{h}_u} \ell_u\| \leq \frac{2\beta_{\max}}{\tau}.$$

Moreover, if there exists an in-cluster hard negative w such that $\text{sim}(\mathbf{h}_u, \mathbf{h}_w) > 0$, then $\nabla_{\mathbf{h}_u} \ell_u \neq 0$, preventing embeddings from collapsing into a trivial solution.

C Details of the Experiments

We evaluate our method on a diverse set of real-world temporal networks, encompassing human proximity, online interactions, and communication patterns. The datasets vary significantly in

terms of scale, skewness, edge density, and temporal granularity, thereby testing the model’s robustness across dynamic structures. All datasets are publicly available and widely used in prior studies on temporal graph learning and dynamic centrality analysis. Table 9 summarizes the key statistics of each dataset. The training graphs are markedly sparser than the test graphs (density 0.075 ± 0.238 vs. 0.133 ± 0.144), indicating a clear structural shift and confirming that our evaluation reflects inductive generalization.

Experiment Setup. For every dataset, we repeated each experiment 30 times, and we reported the mean and the standard deviation of all scores. We set the embedding dimensions of node features, continuous time, and temporal path count to 128, set the number of temporal neighbors sampled per node to 20, and employed a three-layer model trained over 15 epochs. To ensure fair comparison, we repeated all experiments for each deep learning model using three learning rates, 0.1, 0.01, and 0.001, and reported the best mean score. All experiments were conducted on a workstation equipped with an NVIDIA GeForce RTX 3090 GPU, CUDA 12.2, and NVIDIA driver version 535.113.01. Our training dataset consists of 50 real-world temporal networks from Network Repository [27] and Konect [17].

D Additional Experimental Results

D.1 Discretization Sensitivity Analysis

Table 10 compares two discretization strategies, i.e., adaptive thresholding (variance-based) and learnable quantile-based binning, against CLGNN’s default discretization. We report both absolute performance (MAE, Spearman) and relative change Δ . While both alternatives introduce additional flexibility in dynamic range partitioning, they cause substantial degradation across datasets. Adaptive thresholding increases MAE by 24% to 915%, with the most severe case

Table 11: HitsIn10 performance comparison.

Dataset	CLGNN	DrBC	GNN-Bet	DBGNN	TGAT	TATKC	T-Trans	TempGNTK	IIOIwD	ONBRA
sp_hospital	3.0 ± 0.5	0.9 ± 0.6	0.9 ± 0.6	8.0 ± 0.6	1.0 ± 0.3	0.0 ± 0.0	2.4 ± 0.3	1.0 ± 0.3	0.0 ± 0.5	1.0 ± 0.0
sp_workplace	3.1 ± 0.5	0.2 ± 0.2	0.1 ± 0.1	4.0 ± 0.7	0.0 ± 0.0	0.0 ± 0.0	0.4 ± 0.3	1.2 ± 0.3	1.9 ± 0.3	1.0 ± 0.0
Highschool2011	4.3 ± 0.7	2.2 ± 0.4	0.4 ± 0.2	<u>2.0 ± 0.6</u>	0.0 ± 0.0	0.0 ± 0.0	0.3 ± 0.2	0.2 ± 0.2	0.0 ± 0.4	0.0 ± 0.0
Highschool2012	6.3 ± 0.5	1.3 ± 0.6	1.1 ± 0.3	<u>3.0 ± 0.7</u>	0.0 ± 0.0	1.0 ± 0.4	0.2 ± 0.2	0.1 ± 0.1	0.3 ± 0.5	0.0 ± 0.0
haggle	5.2 ± 0.4	0.0 ± 0.0	0.1 ± 0.1	5.0 ± 0.4	0.0 ± 0.0	1.0 ± 0.4	4.0 ± 0.4	0.1 ± 0.1	0.0 ± 0.2	8.0 ± 0.0
ia-reality-call	2.0 ± 0.5	0.0 ± 0.0	0.0 ± 0.0	1.0 ± 0.6	0.0 ± 0.0	1.0 ± 0.4	0.5 ± 0.3	0.0 ± 0.0	0.0 ± 0.4	0.0 ± 0.0
infectious	0.9 ± 0.5	0.2 ± 0.2	0.3 ± 0.3	0.5 ± 0.5	0.0 ± 0.0	0.0 ± 0.0	0.3 ± 0.2	0.1 ± 0.1	0.4 ± 0.5	—
wikiedits-se	0.8 ± 0.4	0.0 ± 0.0	0.2 ± 0.2	1.0 ± 0.6	0.0 ± 0.0	1.0 ± 0.3	0.2 ± 0.2	0.1 ± 0.1	0.3 ± 0.4	—
superuser	<u>0.9 ± 0.4</u>	0.0 ± 0.0	0.3 ± 0.3	OOM	1.0 ± 0.4	0.0 ± 0.0	0.0 ± 0.0	0.1 ± 0.1	0.0 ± 0.4	—

Notes: “—” indicates that, under the same time budget and constraints as CLGNN, ONBRA failed to find a sufficient number of valid temporal paths; OOM indicates the method ran out of memory or exceeded practical resource limits. Best and second-best results are highlighted in **Bold** and underlined, respectively.

Table 12: HitsIn30 performance comparison.

Dataset	CLGNN	DrBC	GNN-Bet	DBGNN	TGAT	TATKC	T-Trans	TempGNTK	IIOIwD	ONBRA
sp_hospital	20.2 ± 0.3	3.8 ± 0.7	3.7 ± 0.3	24.0 ± 0.4	9.0 ± 0.4	10.0 ± 0.5	17.0 ± 0.4	9.0 ± 0.4	8.2 ± 0.4	11.0 ± 0.0
sp_workplace	10.2 ± 0.6	0.9 ± 0.3	7.8 ± 0.4	20.0 ± 0.7	4.0 ± 0.7	9.0 ± 0.5	3.0 ± 0.5	9.0 ± 0.4	<u>10.3 ± 0.3</u>	6.0 ± 0.0
Highschool2011	11.2 ± 0.4	1.8 ± 0.6	1.9 ± 0.5	10.0 ± 0.6	6.0 ± 0.3	<u>11.0 ± 0.4</u>	7.0 ± 0.4	2.0 ± 0.3	<u>0.3 ± 0.4</u>	3.0 ± 0.0
Highschool2012	9.2 ± 0.5	1.1 ± 0.3	3.3 ± 0.3	4.0 ± 0.3	5.0 ± 0.5	10.0 ± 0.6	6.0 ± 0.4	0.7 ± 0.2	0.8 ± 0.2	2.0 ± 0.0
haggle	27.2 ± 0.7	0.0 ± 0.0	0.0 ± 0.0	<u>25.0 ± 0.4</u>	1.0 ± 0.3	8.0 ± 0.3	16.0 ± 0.5	0.3 ± 0.2	0.4 ± 0.5	<u>25.0 ± 0.0</u>
ia-reality-call	13.9 ± 0.5	0.0 ± 0.0	0.0 ± 0.0	<u>11.0 ± 0.5</u>	10.0 ± 0.2	10.0 ± 0.3	9.0 ± 0.4	0.5 ± 0.2	3.2 ± 0.3	<u>4.0 ± 0.0</u>
infectious	4.0 ± 0.6	0.3 ± 0.3	0.0 ± 0.0	12.0 ± 0.4	1.0 ± 0.5	2.0 ± 0.6	2.0 ± 0.4	1.0 ± 0.3	3.4 ± 0.3	—
wikiedits-se	5.1 ± 0.4	0.0 ± 0.0	0.1 ± 0.1	3.0 ± 0.3	5.0 ± 0.3	2.0 ± 0.4	3.0 ± 0.4	2.0 ± 0.3	3.4 ± 0.5	—
superuser	7.0 ± 0.6	0.0 ± 0.0	0.0 ± 0.0	OOM	<u>3.0 ± 0.5</u>	2.0 ± 0.4	0.0 ± 0.0	1.0 ± 0.3	0.5 ± 0.2	—

Table 13: HitsIn50 performance comparison.

Dataset	CLGNN	DrBC	GNN-Bet	DBGNN	TGAT	TATKC	T-Trans	TempGNTK	IIOIwD	ONBRA
sp_hospital	46.2 ± 0.3	17.7 ± 0.7	27.3 ± 0.4	32.0 ± 0.4	31.0 ± 0.7	34.0 ± 0.3	<u>39.0 ± 0.5</u>	21.0 ± 0.5	28.7 ± 0.2	29.0 ± 0.0
sp_workplace	30.8 ± 0.4	7.9 ± 0.6	18.8 ± 0.6	30.0 ± 0.5	19.0 ± 0.7	26.0 ± 0.6	<u>21.0 ± 0.5</u>	21.0 ± 0.5	26.8 ± 0.4	19.0 ± 0.0
Highschool2011	26.0 ± 0.4	12.0 ± 0.4	10.7 ± 0.4	<u>22.0 ± 0.6</u>	<u>25.0 ± 0.5</u>	<u>25.0 ± 0.4</u>	23.0 ± 0.5	7.0 ± 0.4	9.3 ± 0.3	15.0 ± 0.0
Highschool2012	12.0 ± 0.3	9.8 ± 0.4	10.2 ± 0.5	11.0 ± 0.3	<u>14.0 ± 0.5</u>	17.0 ± 0.6	<u>16.0 ± 0.5</u>	4.5 ± 0.3	13.7 ± 0.3	8.0 ± 0.0
haggle	49.8 ± 0.2	2.8 ± 0.5	2.9 ± 0.4	37.0 ± 0.5	18.0 ± 0.7	23.0 ± 0.6	<u>26.0 ± 0.5</u>	1.2 ± 0.4	0.0 ± 0.4	<u>39.0 ± 0.0</u>
ia-reality-call	35.8 ± 0.6	0.0 ± 0.0	0.0 ± 0.0	<u>30.0 ± 0.3</u>	13.0 ± 0.6	20.0 ± 0.5	18.0 ± 0.5	2.0 ± 0.3	5.2 ± 0.3	<u>4.0 ± 0.0</u>
infectious	<u>13.9 ± 0.7</u>	0.8 ± 0.6	0.0 ± 0.0	15.0 ± 0.5	4.0 ± 0.7	9.0 ± 0.2	8.0 ± 0.3	6.0 ± 0.4	10.5 ± 0.5	—
wikiedits-se	12.0 ± 0.3	0.0 ± 0.0	0.0 ± 0.0	<u>10.0 ± 0.5</u>	6.0 ± 0.6	5.0 ± 0.3	6.0 ± 0.4	4.0 ± 0.3	6.0 ± 0.4	—
superuser	12.7 ± 0.4	0.2 ± 0.2	0.0 ± 0.0	OOM	11.0 ± 0.3	<u>12.0 ± 0.6</u>	0.0 ± 0.0	3.0 ± 0.3	2.5 ± 0.2	—

on *haggle* (+915%), while reducing rank correlation by up to 88.7% (*Highschool2013*). Learnable quantile-based binning yields similar or worse instability, with MAE increases of 29% to 750% and *Spearman* drops as large as 84.9% (*Highschool2013*) and 79.8% (*haggle*). These results reveal that additional flexibility comes at the cost of instability, particularly on temporally sparse datasets (e.g., *haggle*, *sp_workplace*). In contrast, CLGNN’s stability-seeking clustering achieves a balanced discretization that preserves temporal smoothness, consistently delivering the best trade-off between accuracy and stability across all datasets.

D.2 HitsIn@K Performance

Tables 11–13 report the HitsIn10, HitsIn30, and HitsIn50 performance of CLGNN compared to the baseline methods. We observe that (i) CLGNN consistently achieves non-zero HitsIn10, 30, 50 across all datasets, whereas DrBC and GNN-Bet fail to retrieve any top-10 nodes in over 40% of the cases, indicating their limitations in dynamic settings. (ii) CLGNN outperforms TempGNTK across all datasets and exceeds TATKC, TGAT, ONBRA, T-Trans, and IIOIwD in 77% ~ 90% of cases. Compared to DBGNN, CLGNN performs

better on 7/9 datasets for HitsIn10, 7/9 for HitsIn30, and 8/9 for HitsIn50. The reasons why DBGNN performs well on some datasets are: (1) it is a transductive model that splits a single graph into training and testing sets, potentially leveraging more direct structural overlap; (2) it constructs 1st to k -th order De Bruijn graphs and performs message passing on all such graphs. This explicitly encodes time-respecting paths and their surrounding structures, which helps capture fine-grained ranking signals. However, the scalability of DBGNN is limited. 1st-order De Bruijn graph is a time-aggregated, weighted graph where each edge represents the count of time-respecting edges between two nodes. In a k -th order De Bruijn graph, each node represents a time-respecting path of length $k-1$ and edges correspond to length- k time-respecting paths. This design incurs exponential space-time complexity, making the construction of higher-order De Bruijn graphs impractical. In contrast, CLGNN is inductive and scalable, capable of handling diverse graph structures and longer-range dependencies without explicitly constructing high-order graphs. Its advantage becomes more pronounced as the top- K threshold increases. For example, on *haggle*, CLGNN achieves 27.2 (HitsIn30) and 49.8 (HitsIn50), significantly outperforming baselines.